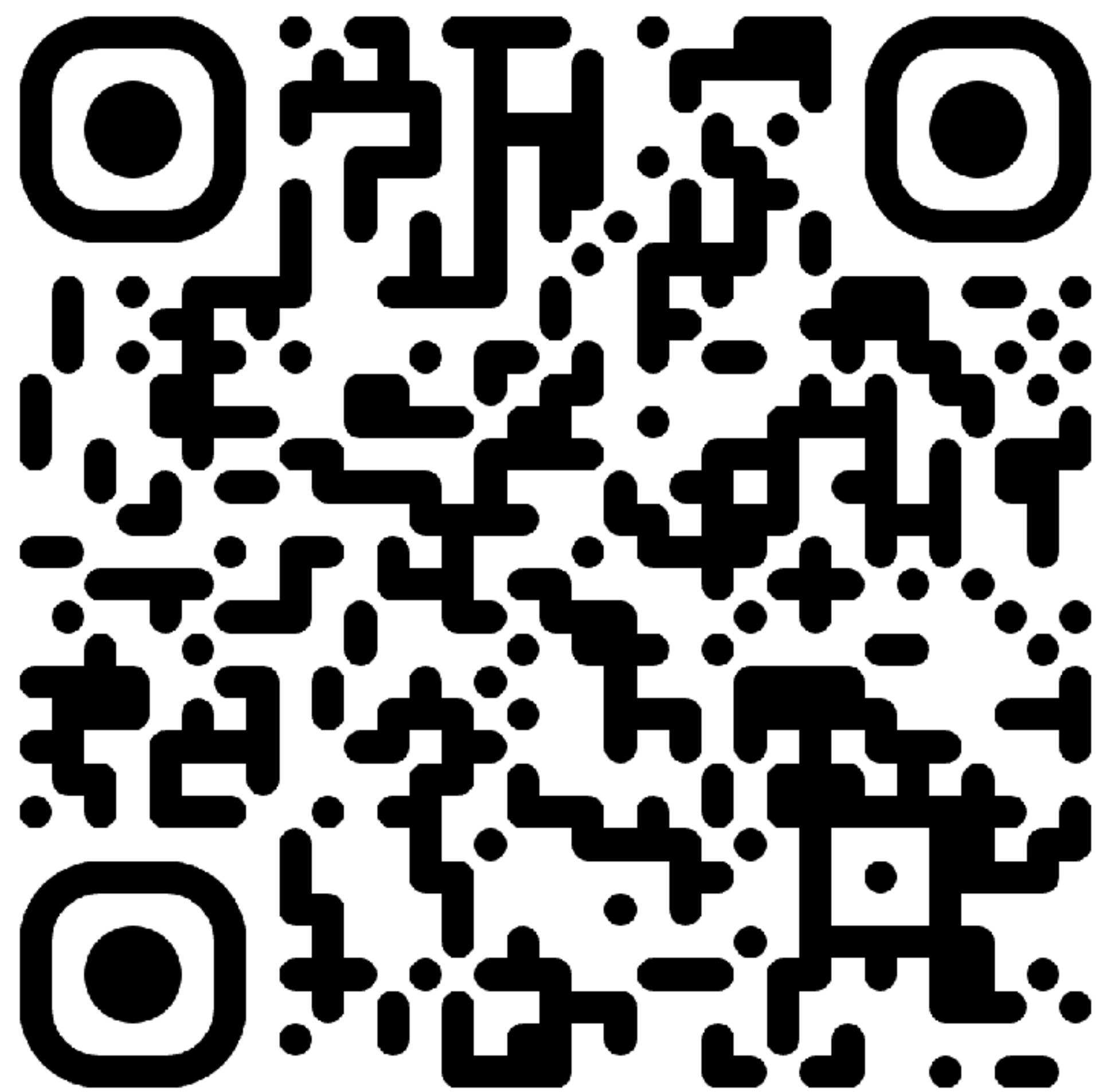


LING2136 / 4136

**Advanced Statistical Methods for
Language Students**

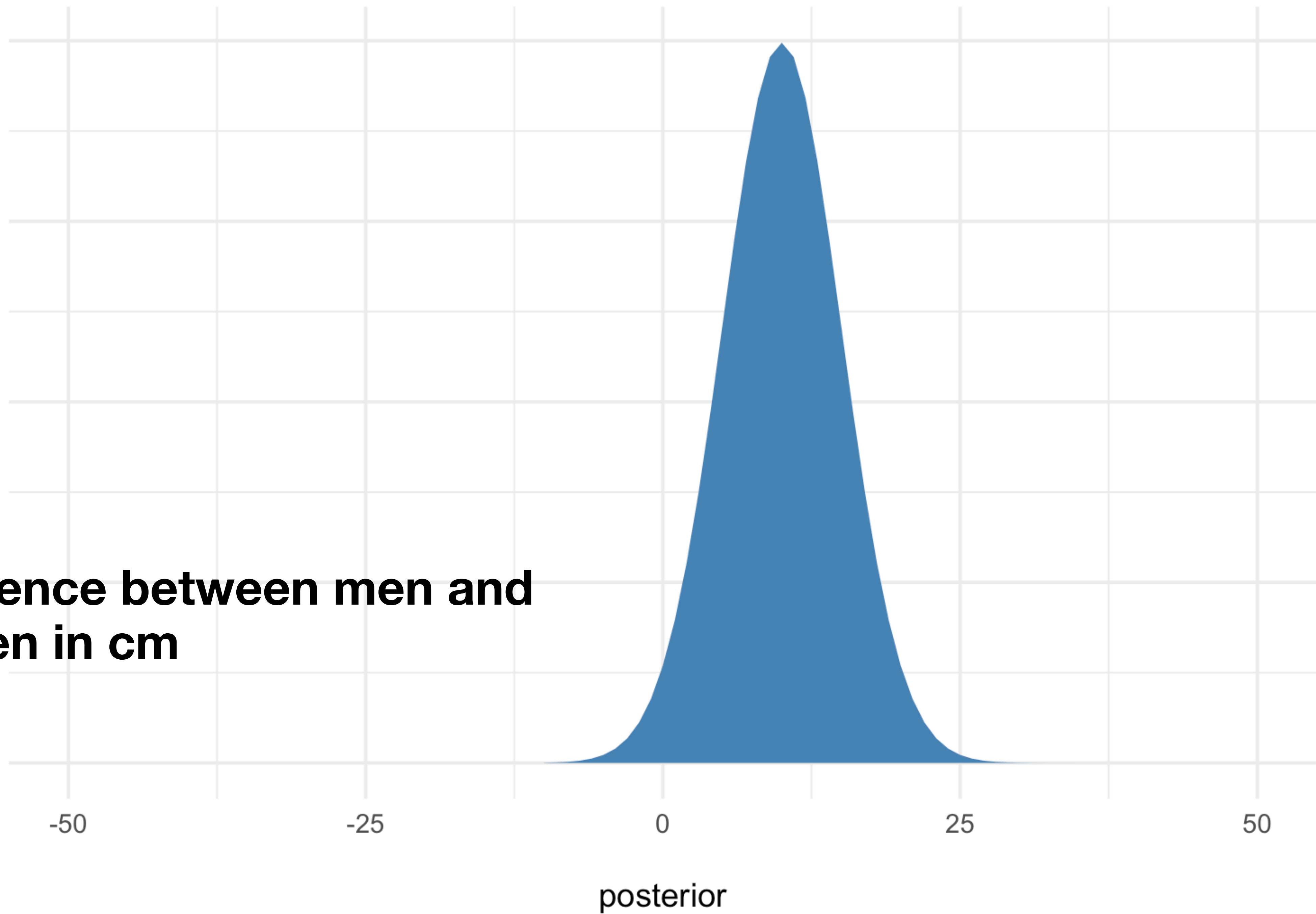
Session-08: inference

Lecturer: Timo Roettger



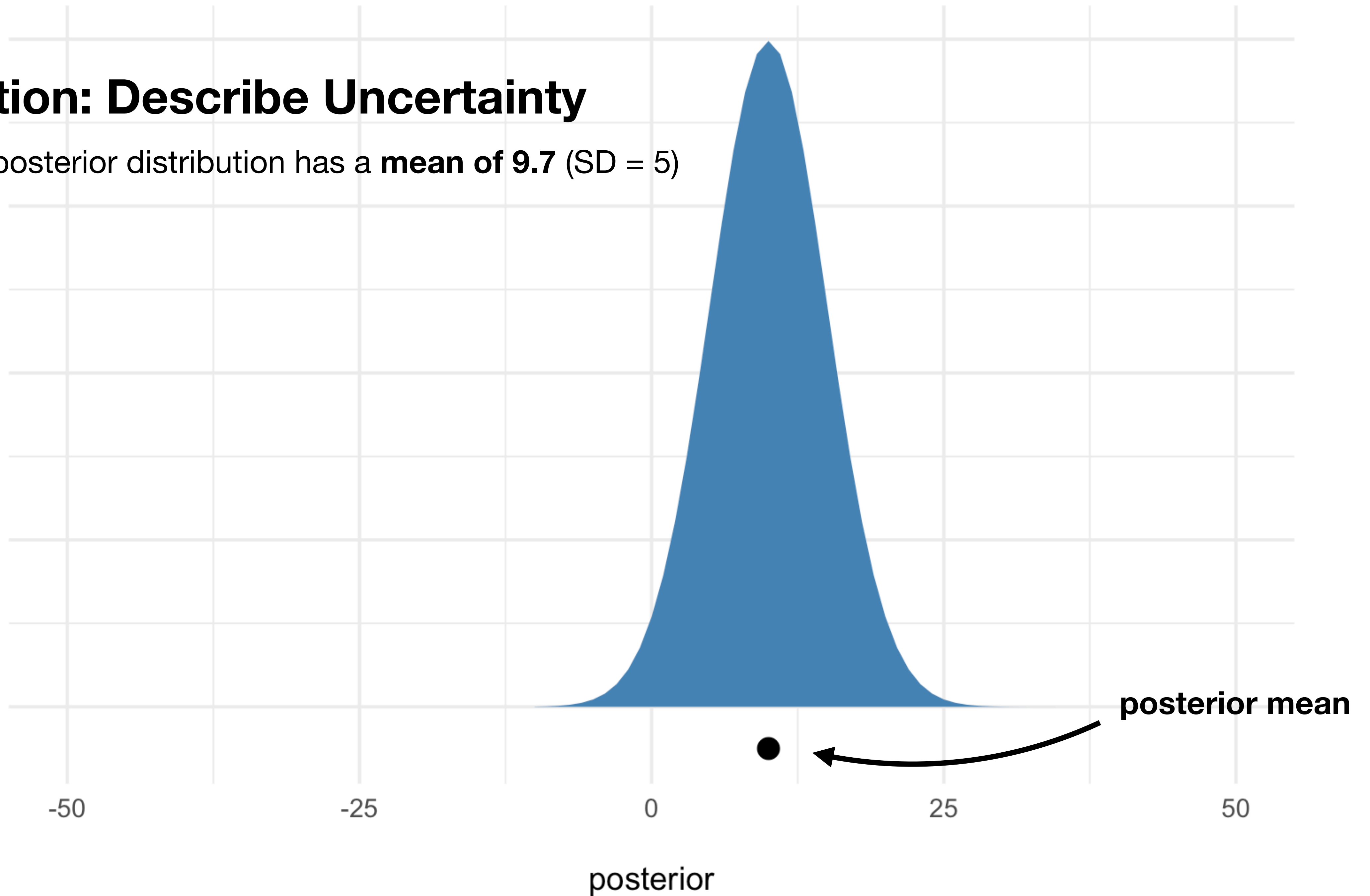
`lm(size ~ sex)`

**Difference between men and
women in cm**



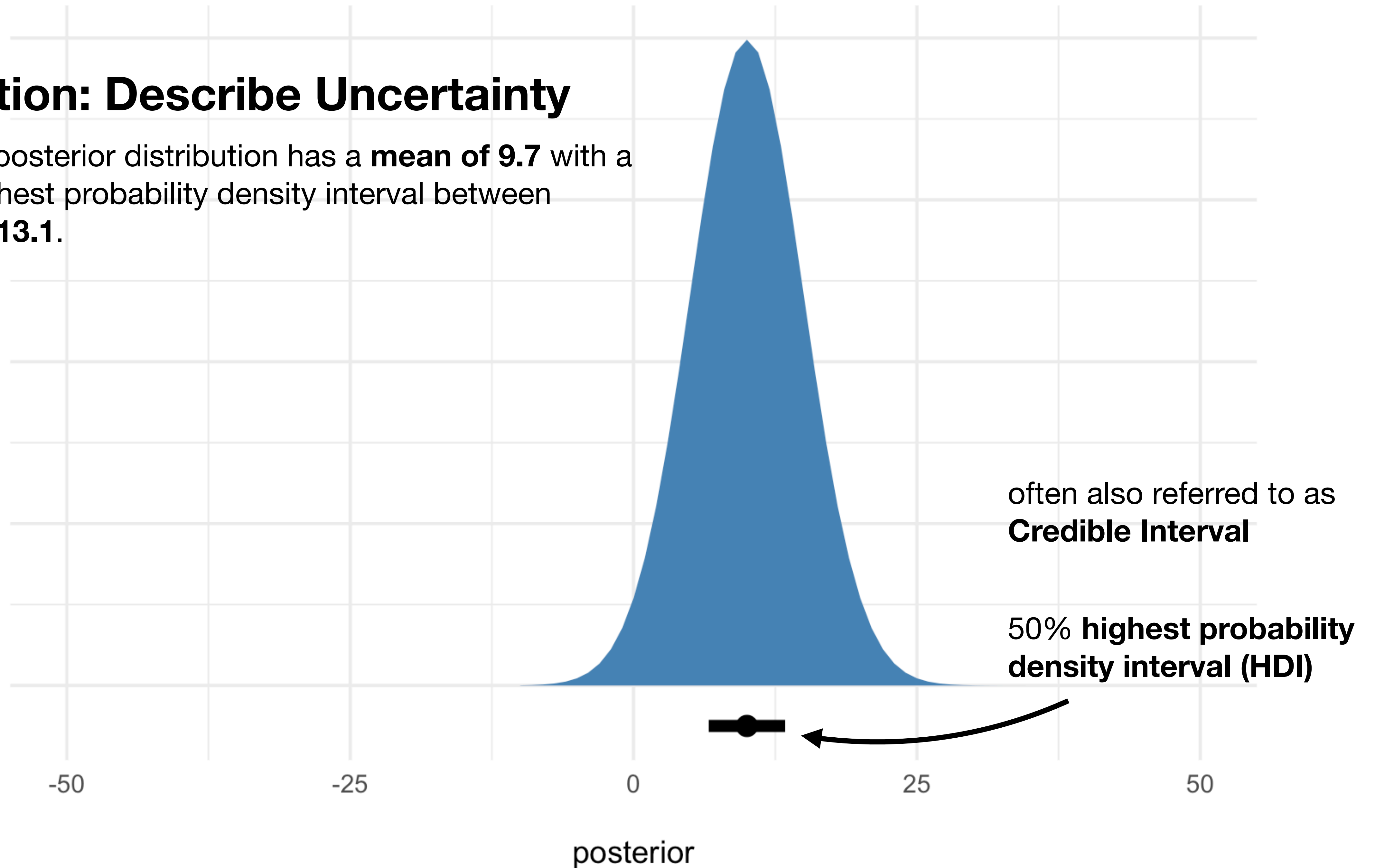
#1 option: Describe Uncertainty

e.g. the posterior distribution has a **mean of 9.7** (SD = 5)



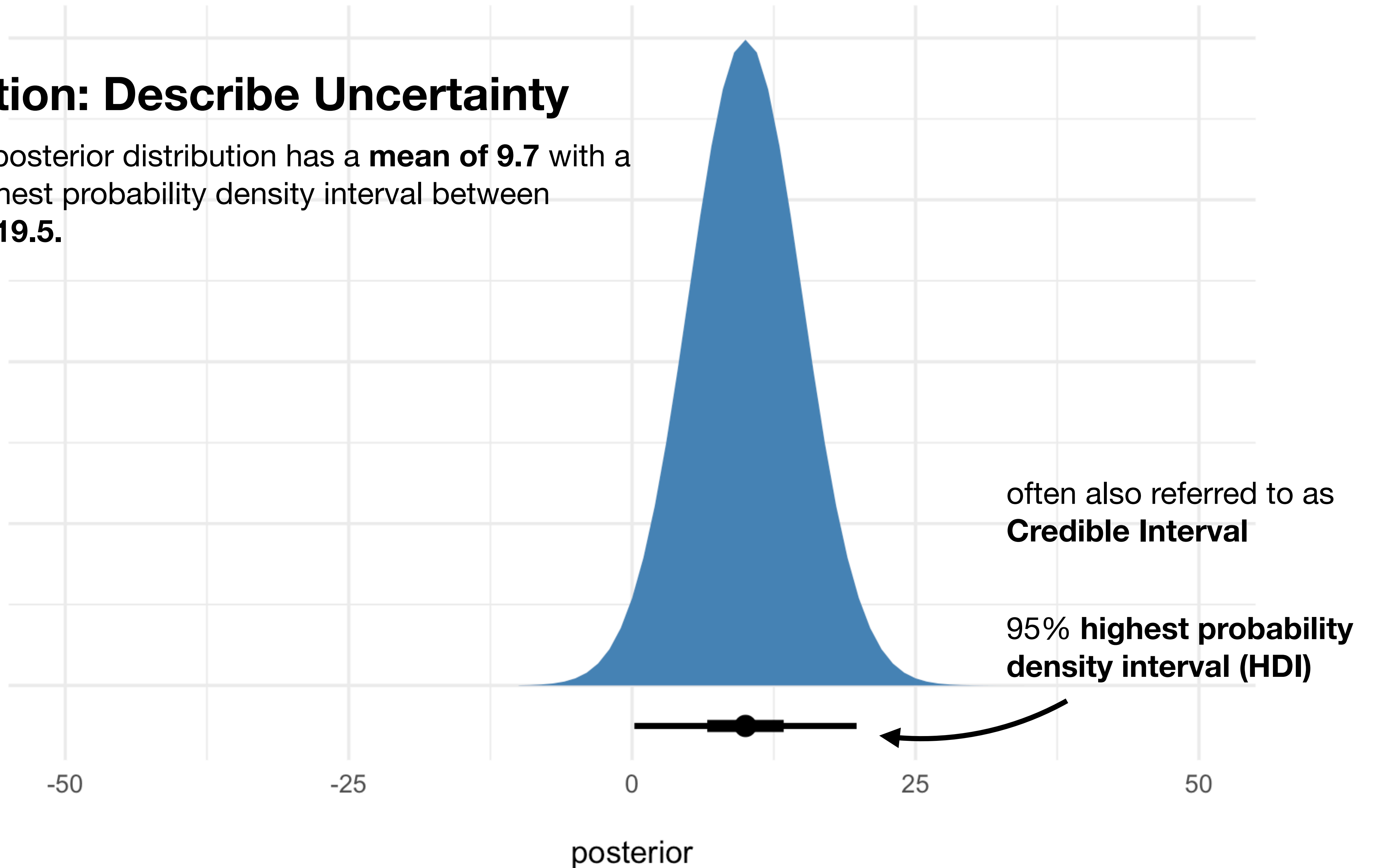
#1 option: Describe Uncertainty

e.g. the posterior distribution has a **mean of 9.7** with a **50% highest probability density interval between 6.3 and 13.1**.



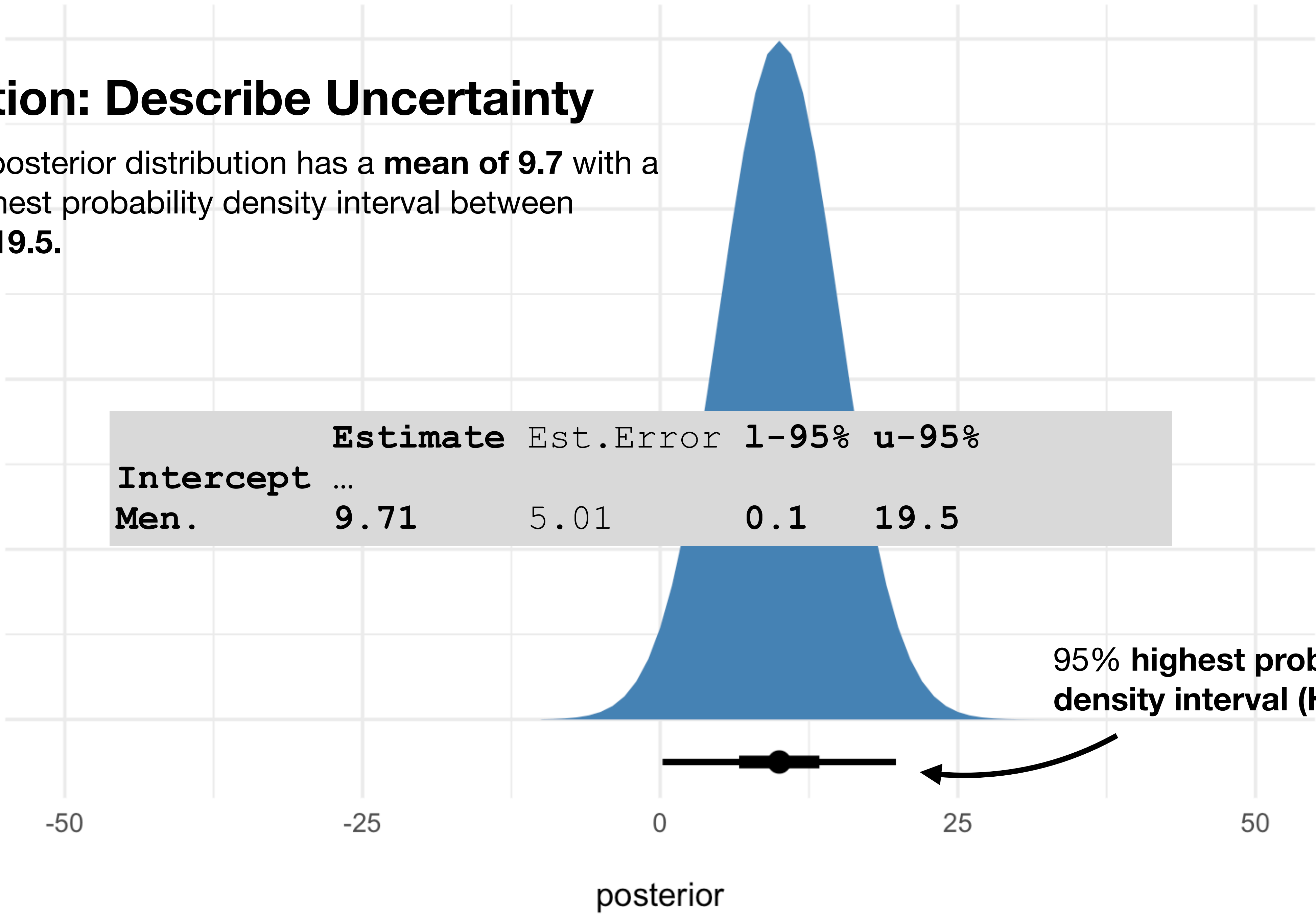
#1 option: Describe Uncertainty

e.g. the posterior distribution has a **mean of 9.7** with a **95%** highest probability density interval between **0.1 and 19.5**.

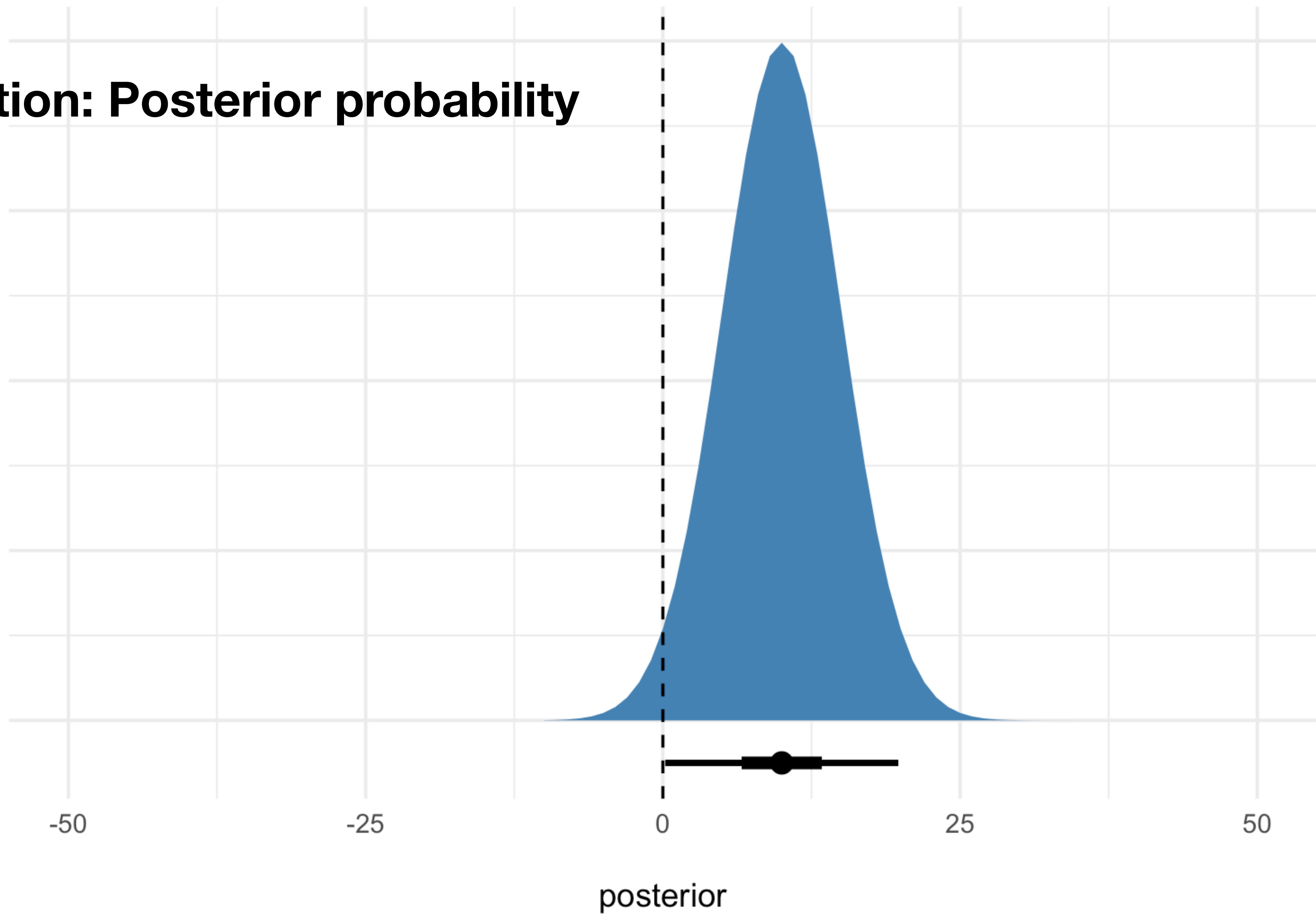


#1 option: Describe Uncertainty

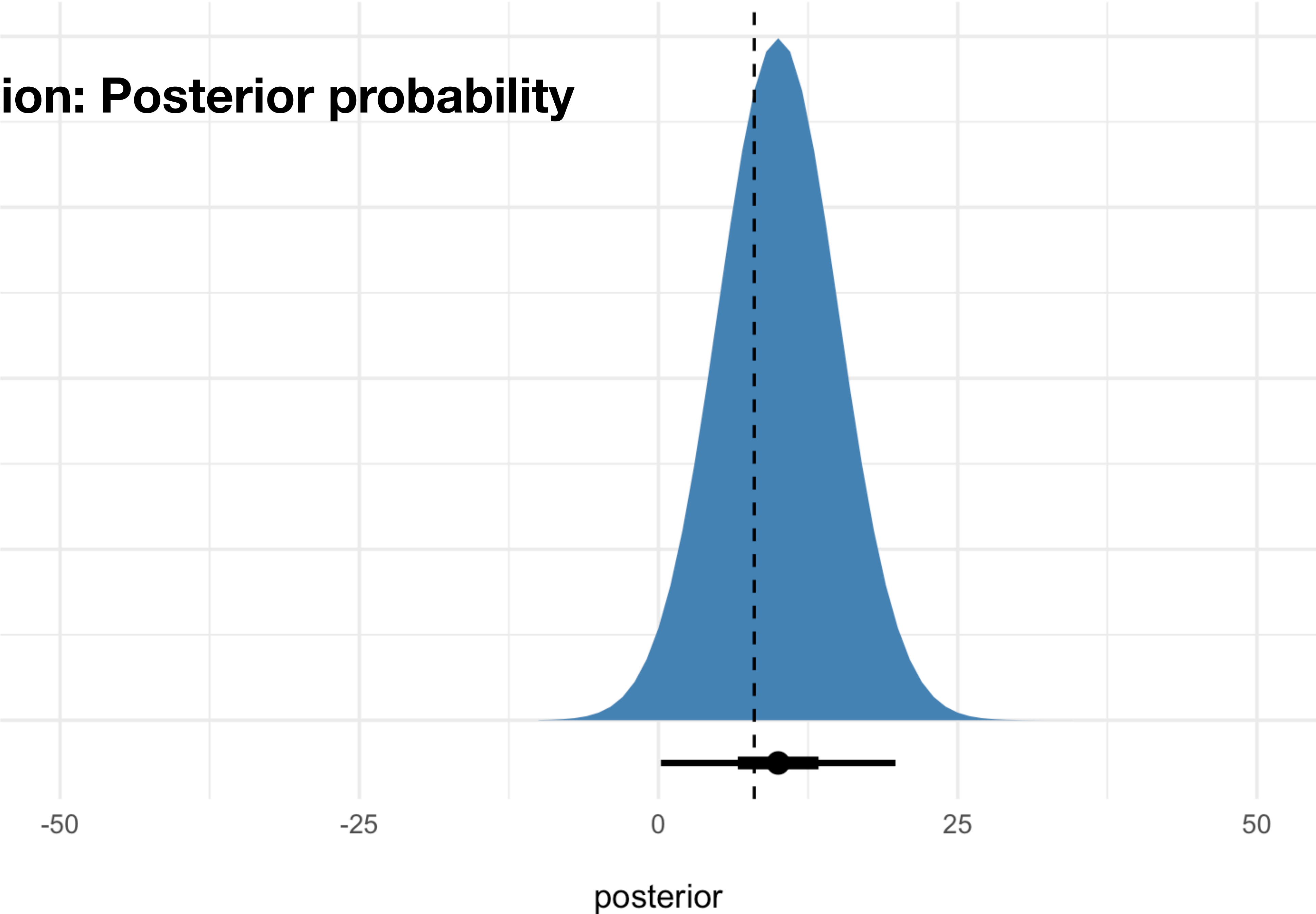
e.g. the posterior distribution has a **mean of 9.7** with a **95%** highest probability density interval between **0.1** and **19.5**.



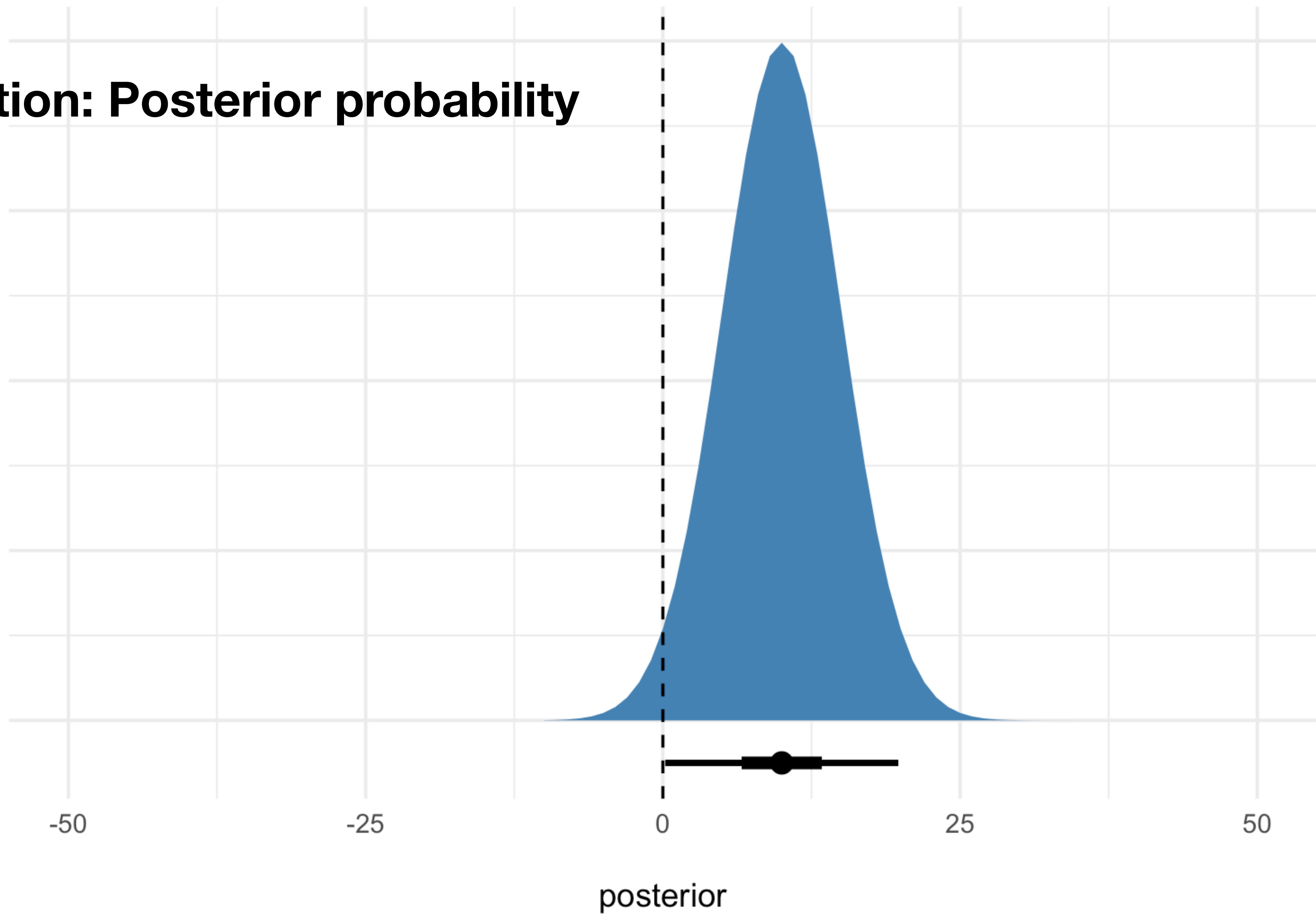
#2 option: Posterior probability



#2 option: Posterior probability



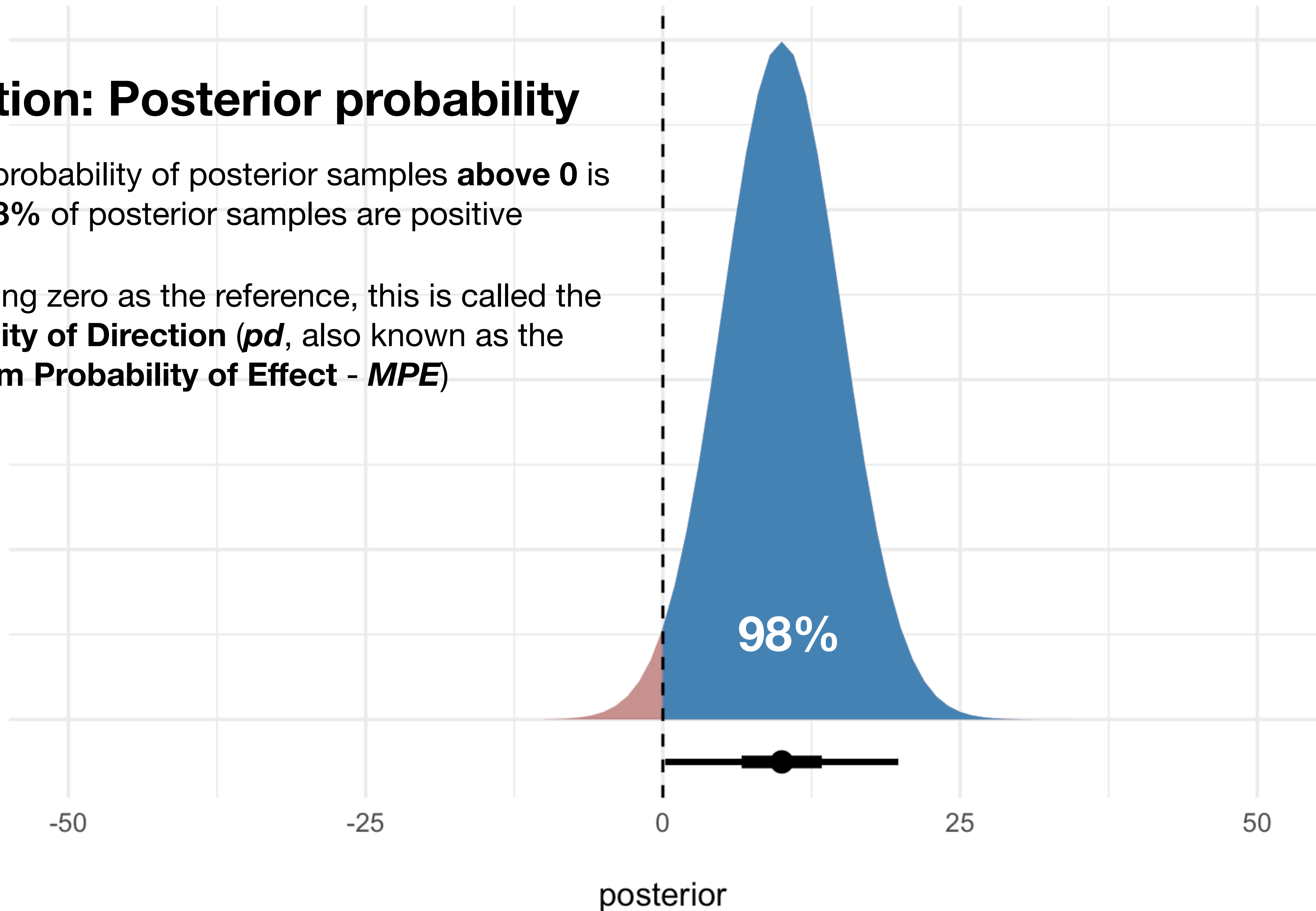
#2 option: Posterior probability



#2 option: Posterior probability

e.g. the probability of posterior samples **above 0** is **0.98 ~ 98%** of posterior samples are positive

when using zero as the reference, this is called the **Probability of Direction (*pd*)**, also known as the **Maximum Probability of Effect - *MPE***)



#2 option: Posterior probability

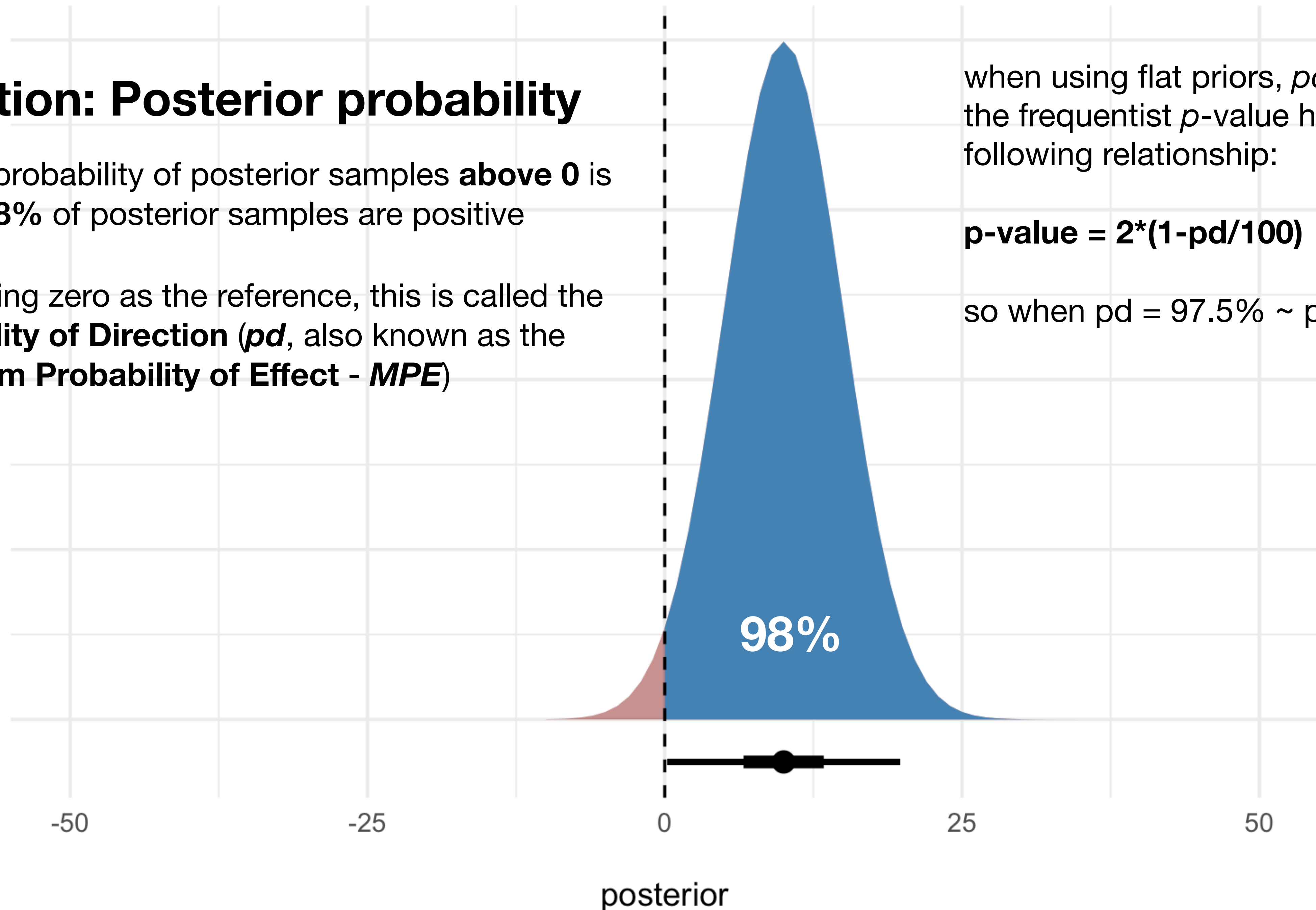
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when using zero as the reference, this is called the **Probability of Direction (*pd*)**, also known as the **Maximum Probability of Effect - *MPE***)

when using flat priors, *pd* and the frequentist *p*-value have the following relationship:

$$\text{p-value} = 2 \cdot (1 - \text{pd} / 100)$$

so when $\text{pd} = 97.5\% \sim p = 0.05$



#2 option: Posterior probability

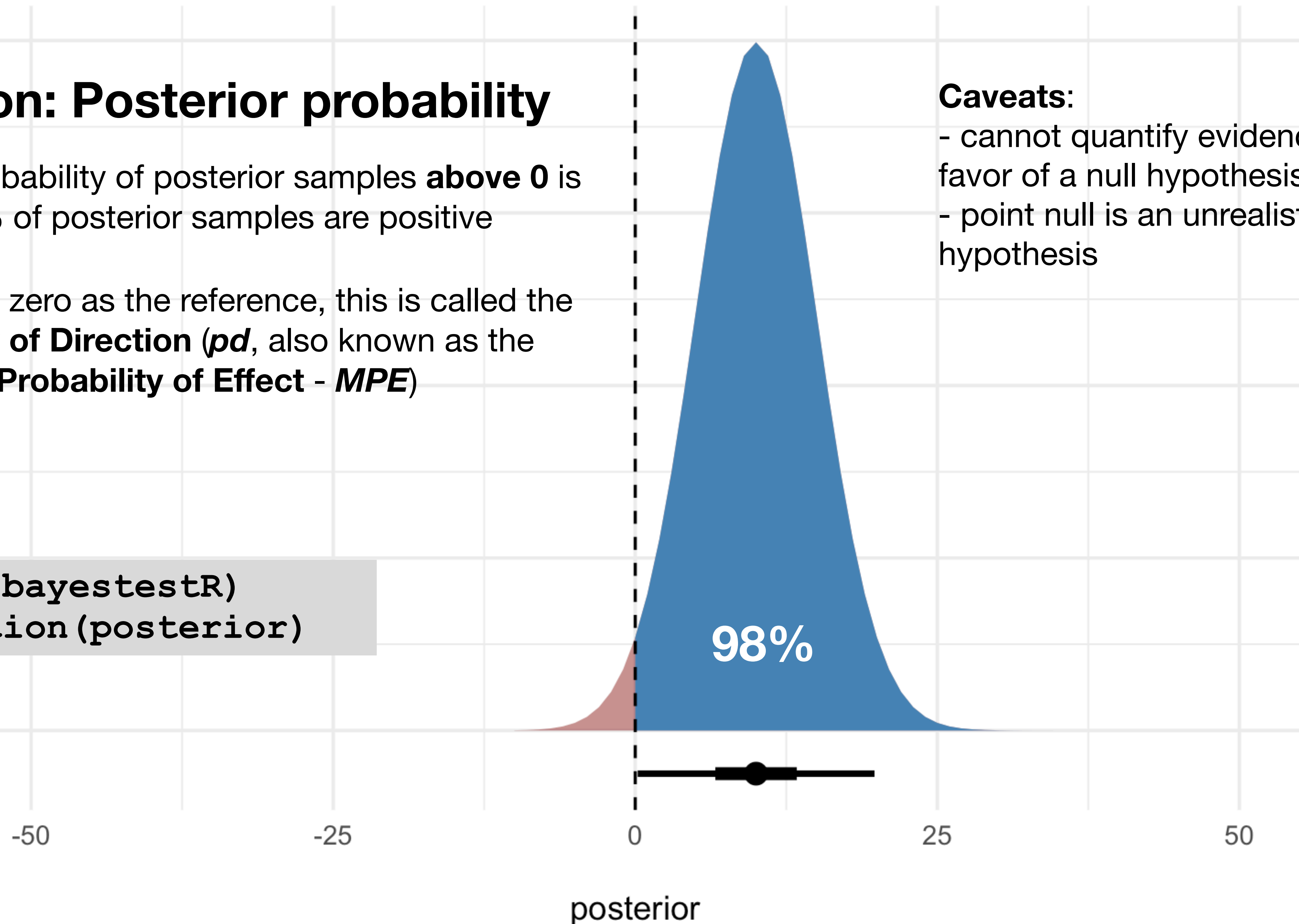
e.g. the probability of posterior samples **above 0** is **0.98 ~ 98%** of posterior samples are positive

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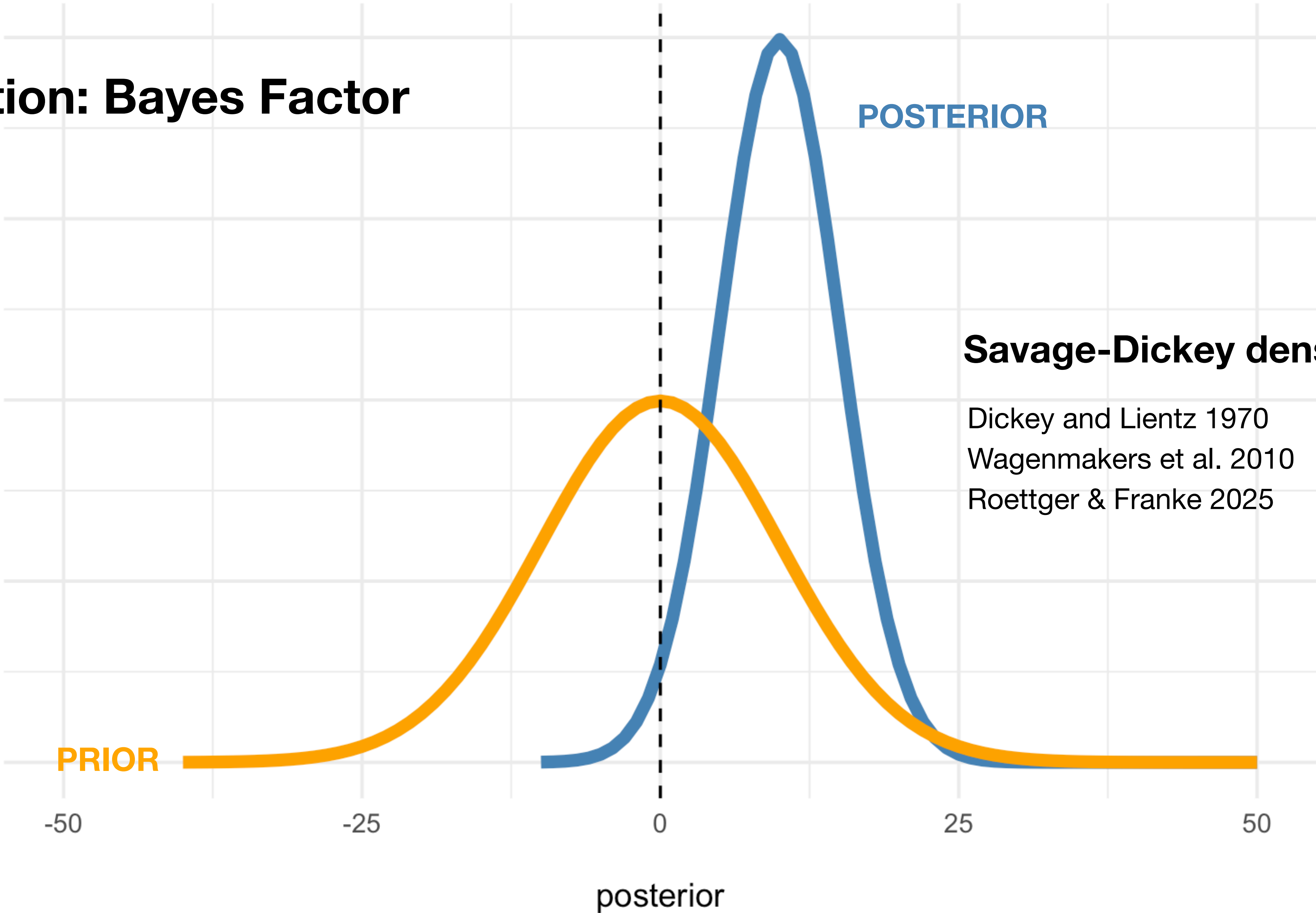
```
library(bayestestR)  
p_direction(posterior)
```

Caveats:

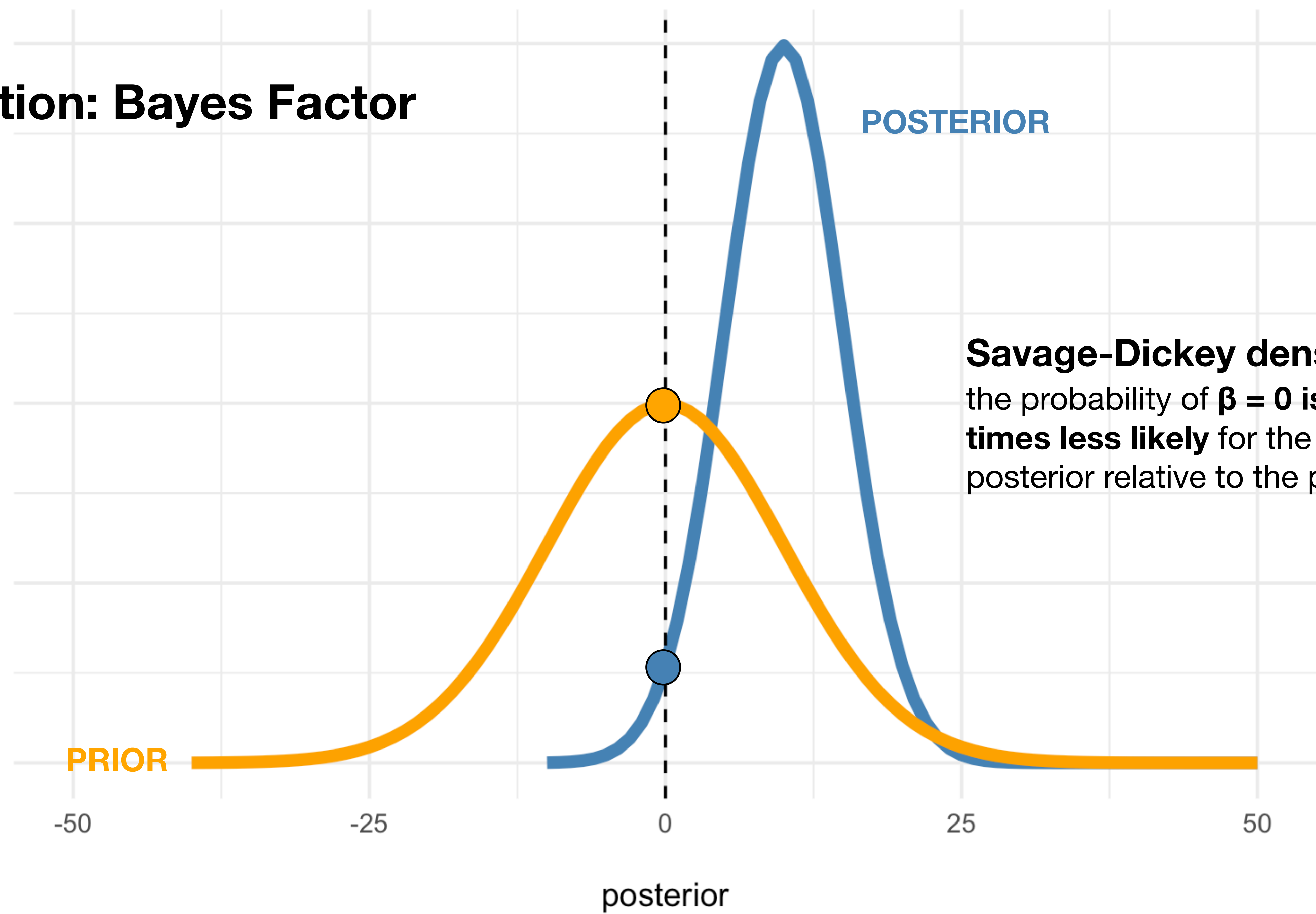
- cannot quantify evidence in favor of a null hypothesis
- point null is an unrealistic null hypothesis



#3 option: Bayes Factor



#3 option: Bayes Factor



#3 option: Bayes Factor

against any value based on prior and posterior samples of a parameter. It indicates the degree by which the mass of the posterior distribution has shifted further away from / closer to the specified value relative to the prior distribution, thus indicating if the specified value has become less/more likely given the observed data.

```
library(bayestestR)
bayesfactor_parameters(posterior,
  prior,
  null = 0)
```

PRIOR

POSTERIOR

Savage-Dickey density ratio

the probability of $\beta = 0$ is **3.7 times less likely** for the posterior relative to the prior

-50

-25

0

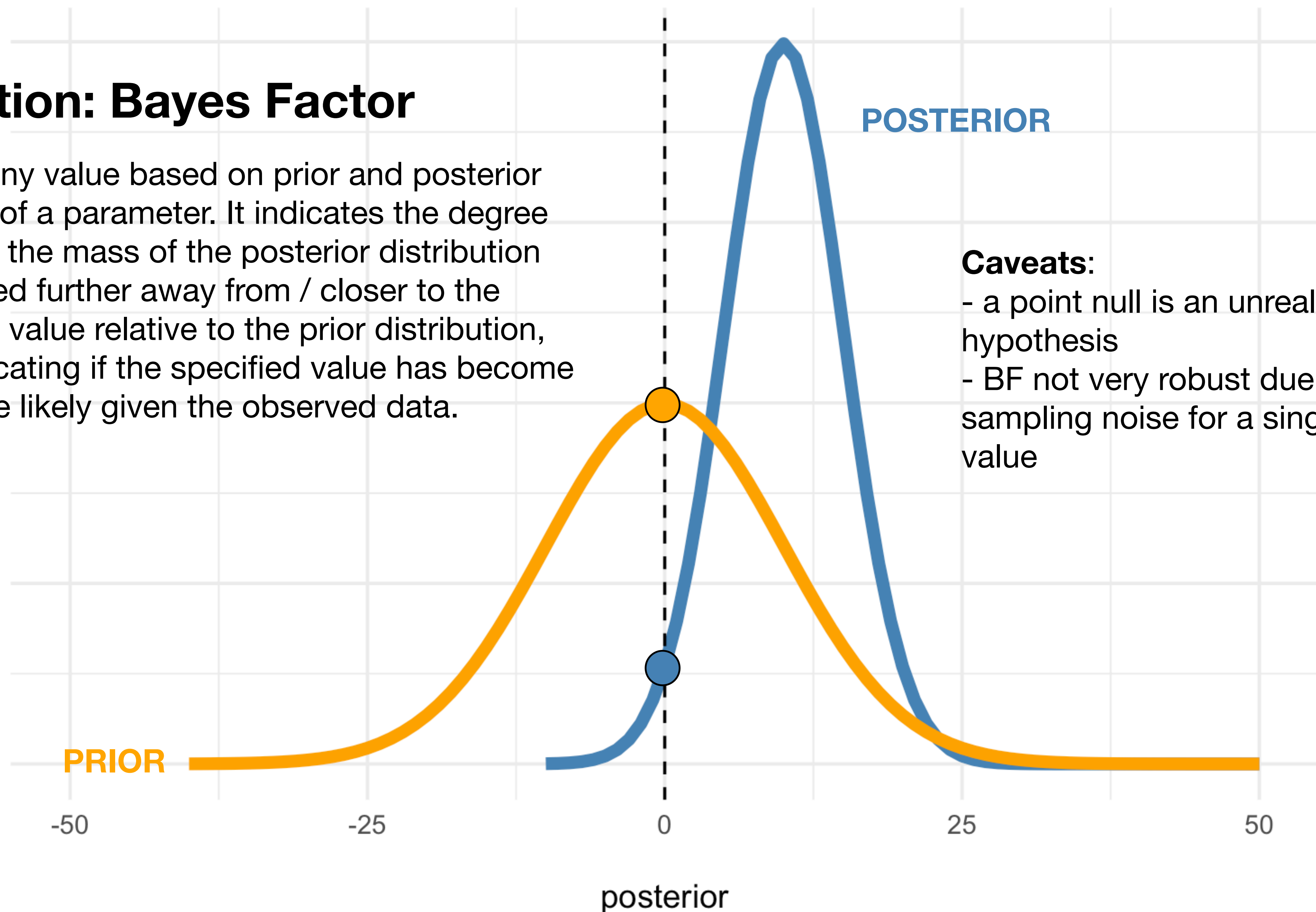
25

50

posterior

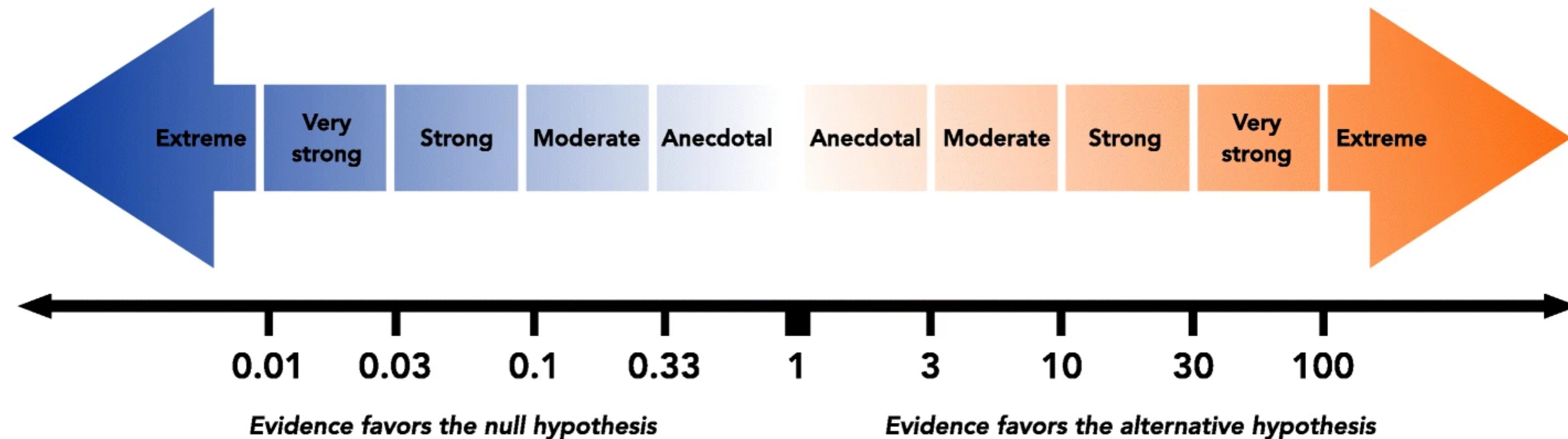
#3 option: Bayes Factor

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Bayes Factor Interpretation Scheme

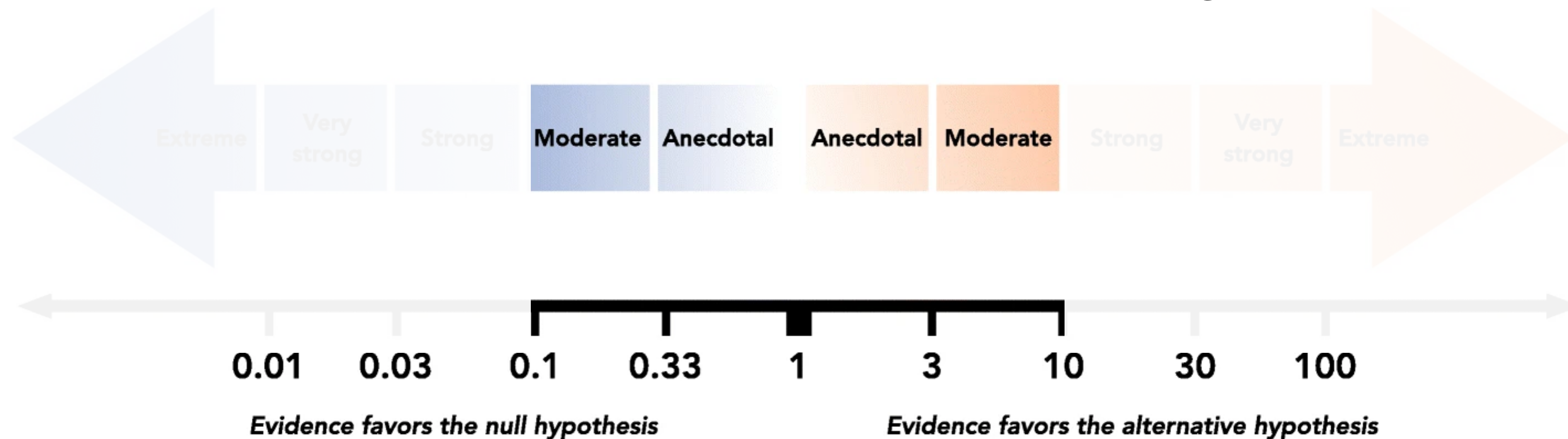
Lee & Wagenmakers (2014)



Bayes Factor Interpretation Scheme

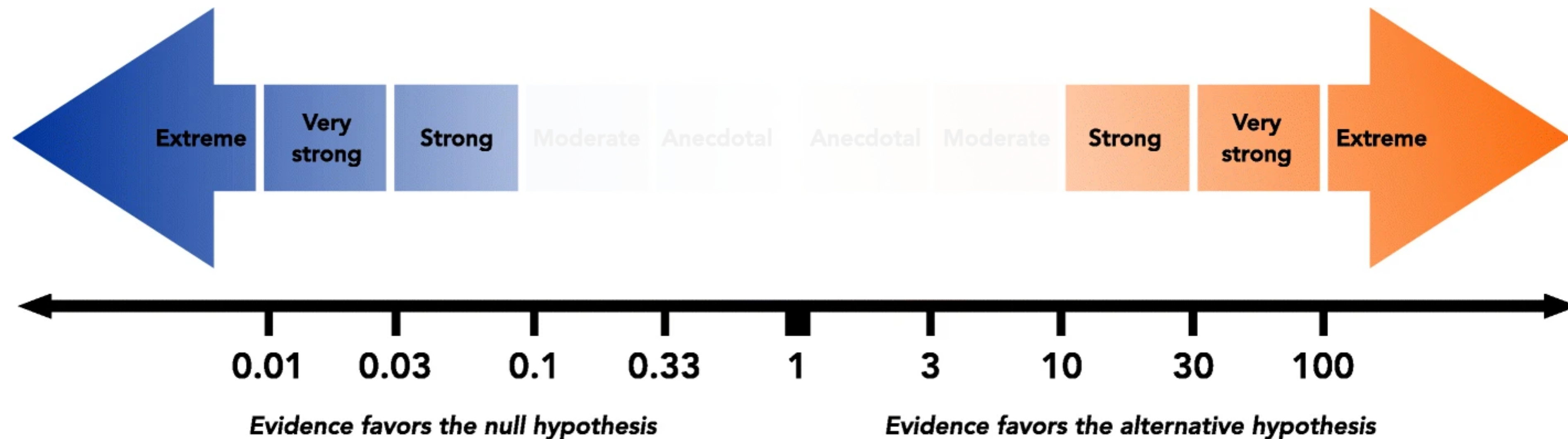
Lee & Wagenmakers (2014)

should be rather careful within this range



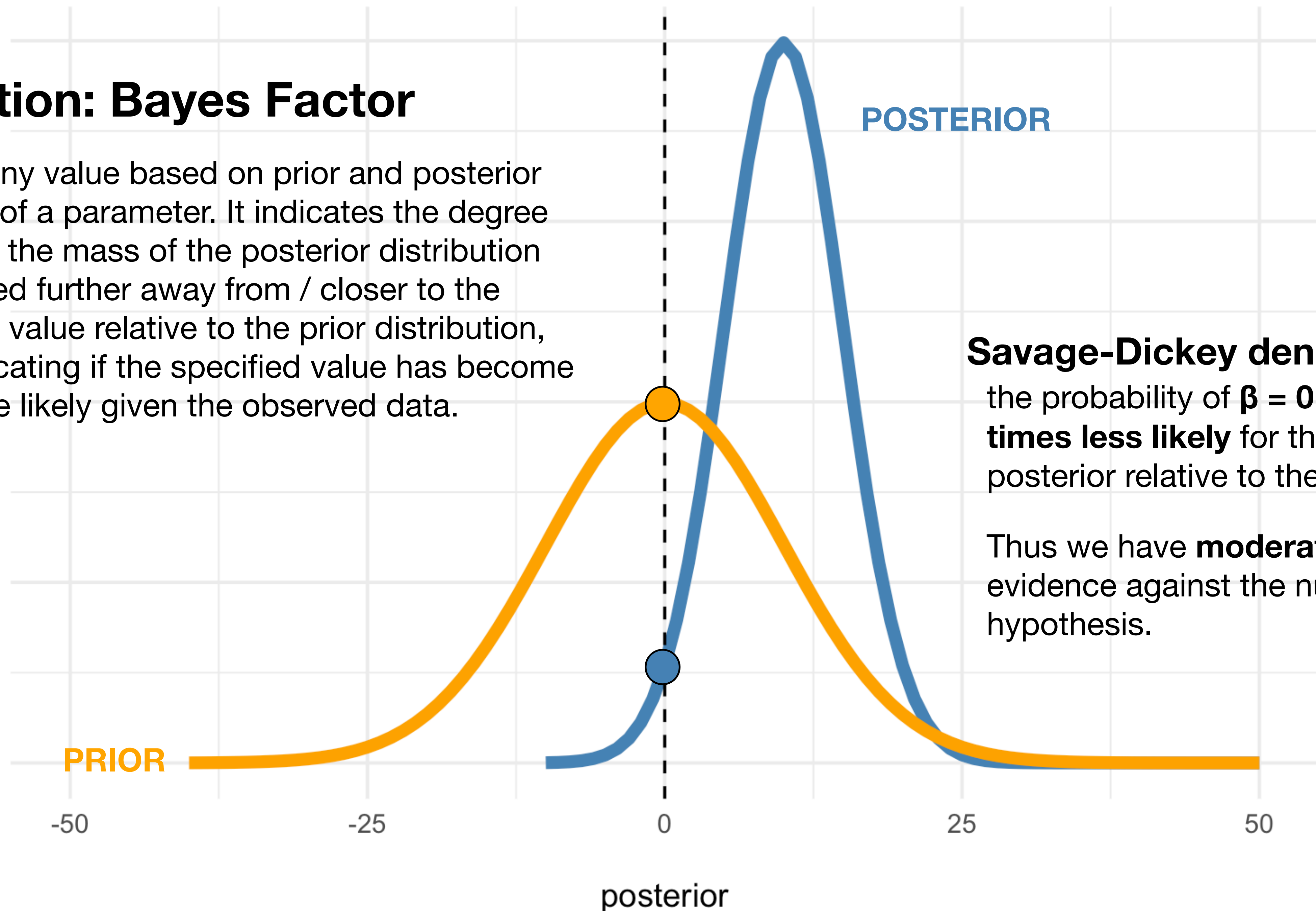
Bayes Factor Interpretation Scheme

Lee & Wagenmakers (2014)



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against any value based on prior and posterior samples of a parameter. It indicates the degree by which the mass of the posterior distribution has shifted further away from / closer to the specified value relative to the prior distribution, thus indicating if the specified value has become less/more likely given the observed data.



Savage-Dickey density ratio

the probability of $\beta = 0$ is **3.7 times less likely** for the posterior relative to the prior.

Thus we have **moderate** evidence against the null hypothesis.